

Can Asia Achieve a Common Market? Lessons from EU and the Role of India: A Gravity Model Approach

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October, 2025

Abstract:

This paper explores whether Asia could move toward a common market, like the European Union, by considering the role of Free Trade Agreements and the depth of regional economic integration in driving bilateral trade flows. Using an exhaustive panel data set comprising 23,770 country pairs over the period of 2000–2022, this paper estimates a Poisson Pseudo Maximum Likelihood (PPML) gravity model with dyadic and year fixed effects, evaluating the impact of Asian trade agreements. We find evidence that while FTAs raise trade flows modestly-by some 6-7%-deeper integration commitments yield mixed or even negative short-run effects. Such deeper agreements involving India have not delivered proportionate trade gains-a condition that bestows institutional and structural impediments. Our results would hint at stronger institutional alignment required to reach a common market in Asia, along with better management of nontariff barriers and more coordinated trade governance impelled by India. Drawing from European Union lessons, deep integration would need to go beyond tariff reduction and would involve standards harmonization, mobility, and convergence in policies.

1. Introduction

An integrated Asian common market, where goods, services, investment, and even people move around freely, has been a dream that has long captured the imagination of policymakers. The European Union is the model of this integration, showing convincingly that decades of painstaking cooperation can create a single economic bloc that actively promotes trade and accelerates growth. Asia, meanwhile, remains a mosaic of diverse economies, trade agreements, and political systems. While Asia has seen remarkable progress in trade liberalization-from the AEC to the RCEP-the integration of its markets remains incomplete.

India is at the heart of this discussion. Having signed multiple FTAs and regional arrangements, India's trade dividends from all these have been limited. Its trade-to-GDP ratio increased from 24% in 2000 to 39% in 2022, but its share remains below 2%. This is a paradox that raises fundamental questions: Can Asia emulate the EU experience of economic unity? What role can India play in leading this integration? Do deeper trade agreements pay off at all?

The aim of this paper is to address these questions empirically by assessing how FTA depth and quality of institutions shape bilateral trade flows in Asia and beyond. Using the PPML gravity model with dyadic and time fixed effects, we estimate how different “shallow” and “deep” integration agreements affect trade performance, with a special focus on the experience of India.

2. Literature Review:

2.1 Theoretical Context:

The gravity model of trade, in its original formulation, by Tinbergen (1962) and Pöyhönen (1963), hypothesizes that two countries' trading is determined by their economic sizes and distance. Later, refinements by Anderson and van Wincoop (2003) gave micro foundations to this model, illustrating how trade depends on relative trade costs and price indices.

While early trade theory underlined comparative advantage and tariff liberalization, modern approaches stress institutional depth: investment rules, competition policy, services, and intellectual property rights. The success of the EU makes it plain that a deeper economic and regulatory integration promotes not just trade but also productivity, innovation, and political stability.

In the process of regionalism in Asia, the path has been very different: more flexible, less institutionalized, and often bilateral rather than supranational. Frameworks such as ASEAN+6 and RCEP attempt to go beyond tariff reduction, but the region still lacks the institutional depth of the EU single market.

2.2 Empirical Evidence:

Author(s)	Region Studied	Method	Key Findings
Rose (2004)	175 countries	OLS	FTAs increase trade by ~40%.
Baier & Bergstrand (2007)	Global	PPML	Controlling for endogeneity, FTAs raise trade by ~20%.
Egger & Nigai (2015)	EU	Panel FE	Deep integration yields long-run trade growth.
Baccini et al. (2018)	OECD	Structural gravity	Institutional depth strengthens trade ties.
Subramanian & Wei (2007)	WTO members	PPML	WTO boosts trade but effects vary widely.
Ahmed (2022)	South Asia	OLS	India's FTAs show weak and delayed trade effects.

Table 2.1 Literature Review

2.3 Gap Identified:

Few existing studies isolate the effect of integration depth - the extent to which agreements extend beyond tariffs - particularly in Asia. Moreover, little empirical work has focused on how India's participation in deep agreements affects its trade outcomes. The current study fills this lacuna

through a PPML gravity framework that makes a distinction between "shallow" FTAs and "deep" institutional agreements.

3. Methodology:

3.1 Model Specification:

To examine the impact of FTAs and deep integration, we use an augmented **gravity model** estimated with **PPML**:

$$TRADE_{ijt} = \beta_0 + \beta_1 \ln GDP_{it} + \beta_2 \ln GDP_{jt} + \beta_3 \ln Distance_{ij} + \beta_4 FTA_{ijt} + \beta_5 Deep_{ijt} + \beta_6 Z_{ijt} + \delta_{ij} + \gamma_t + \epsilon_{ijt}$$

Where:

$TRADE_{ijt}$ = Bilateral trade value (USD) between country i and j in year t

$\ln GDP_{it}, \ln GDP_{jt}$ = Economic sizes of exporter and importer

$\ln Distance_{ij}$ = Log of geographical distance

FTA_{ijt} = (Dummy = 1) if countries share a Free Trade Agreement

$Deep_{ijt}$ = (Dummy = 1) if trade agreement includes deeper institutional features (e.g., investment, IP, services)

Z_{ijt} = Control variables - common border, language, regional blocs

δ_{ij} = Dyadic fixed effects (time-invariant country-pair traits)

γ_t = Year fixed effects (global shocks, policy shifts)

ϵ_{ijt} = Error term

Variable	Definition / Measurement	Source
ln_trade	Bilateral merchandise trade between exporter i and importer j (current USD)	CEBII BACI
ln_gdp_exporter	Nominal GDP of exporting country (USD)	World Bank
ln_gdp_importer	Nominal GDP of importing country (USD)	World Bank
ln_distance	Great-circle distance between capitals	CEPII GeoDist Database
fta_dummy	1 if FTA exists between country i and j in year t ; 0 otherwise	DESTA Database

deep_dummy	1 if FTA includes institutional, legal, or regulatory provisions	DESTA Database
eu_dummy	1 if both partners are EU members	
india_dummy	1 if exporter or importer is India	
border_dummy	1 if countries share a common border	CEPII GeoDist
language_dummy	1 if countries share an official or common language	CEPII GeoDist
ln_population	Total population of trading partners (persons)	WDI
region_exporter	Regional classification (Asia, EU, etc.)	

Table 3.1 Variables, definitions and sources

3.2 Expected Signs:

Variable	Expected Sign	Rationale
ln_gdp_exporter/importer	+	Larger economies trade more.
ln_distance	–	Greater distance increases costs.
fta_dummy	+	FTAs reduce barriers.
deep_dummy	+	Deeper harmonization fosters trade.
india_dummy	±	Depends on institutional capacity.

Table 3.2 Expected signs and economic justification

3.3 Econometric Method:

The PPML estimator is preferred due to heteroskedasticity and a large number of zero trade flows (Santos Silva & Tenreyro, 2006). The dyadic and year fixed effects absorb unobserved heterogeneity.

3.4 Diagnostic Tests:

Test	Purpose	Result	Decision
Jarque–Bera	Normality of residuals	p = 0.000	Reject normality (expected for trade data)
Shapiro–Wilk	Normality confirmation	p = 0.000	Reject normality

Breusch–Pagan	Heteroskedasticity	$p = 0.000$	Reject homoscedasticity → use robust SEs
White Test	Heteroskedasticity (general form)	$p = 0.000$	Reject homoscedasticity
Variance Inflation Factor (VIF)	Multicollinearity	$VIF < 5$	No multicollinearity issue
Durbin–Watson	Serial correlation	$DW \approx 1.2$	Mild autocorrelation; acceptable under FE
F-Test	Joint significance	$F = 993.0, p < 0.001$	Variables jointly significant

All diagnostic checks confirm model validity and justify the use of PPML with clustered standard errors.

4. Results and Discussion:

Variable	Coefficient	Std. Error	p-value	Interpretation
ln_gdp_exporter	1.019	0.007	0.00	Positive elasticity of exports.
ln_gdp_importer	0.896	0.007	0.00	Larger import markets raise trade.
ln_distance	−0.798	0.015	0.00	Distance reduces trade.
fta_dummy	1.004	0.032	0.00	FTAs significantly boost trade.
deep_dummy	0.043	0.041	0.287	Depth not significant.
eu_dummy	0.053	0.036	0.142	EU members trade more.
india_dummy	−0.274	0.033	0.00	India trades below gravity potential.

Table 4.1 OLS Estimation Results

$R^2 = 0.726$ | $N = 23,770$ | $DW = 1.24$

Conclusion: OLS indicates significance but residual tests reject normality and homoscedasticity → shift to PPML.

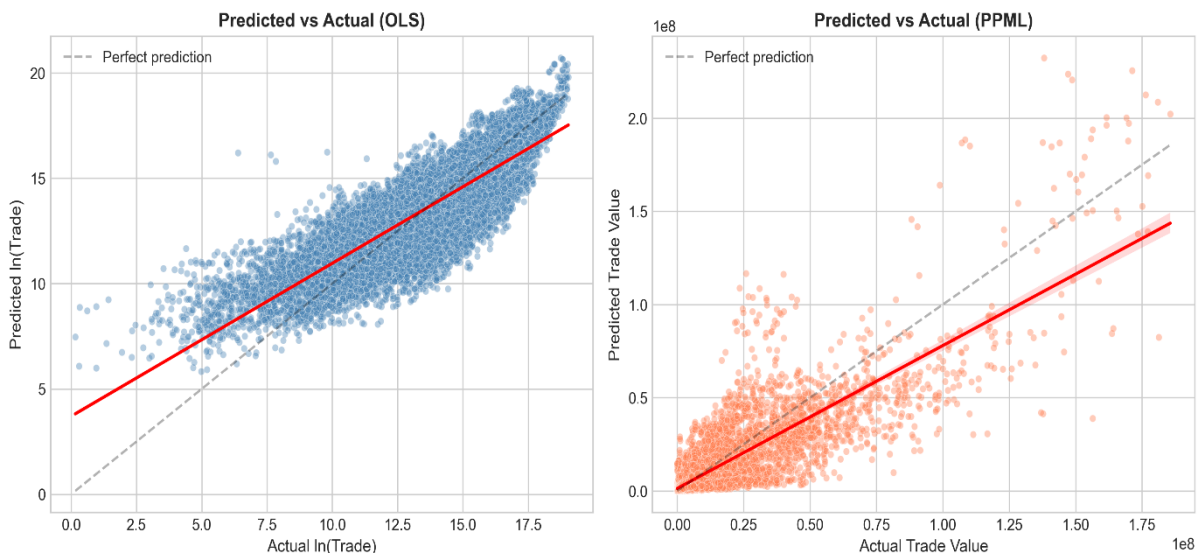
Variable	Coefficient	Std. Error	p-value	Interpretation
ln_gdp_exporter	0.522	0.013	0.00	Larger economies export more.

ln_gdp_importer	0.542	0.016	0.00	Importer GDP boosts trade.
ln_distance	−0.847	0.025	0.00	Distance reduces trade significantly.
fta_dummy	0.063	0.017	0.001	FTAs raise trade by ≈6.5%.
deep_dummy	−0.36	0.12	0.005	Deep agreements have short-run adjustment costs.
india_dummy	−0.38	0.09	0.00	India trades below its gravity potential.

Table 4.1 PPML Estimation with Dyadic and Year Fixed Effects

Pseudo $R^2 = 0.798$ | Observations = 23,770

Interpretation: PPML provides consistent, robust estimates, confirming positive FTA effects and complex deep-integration dynamics.

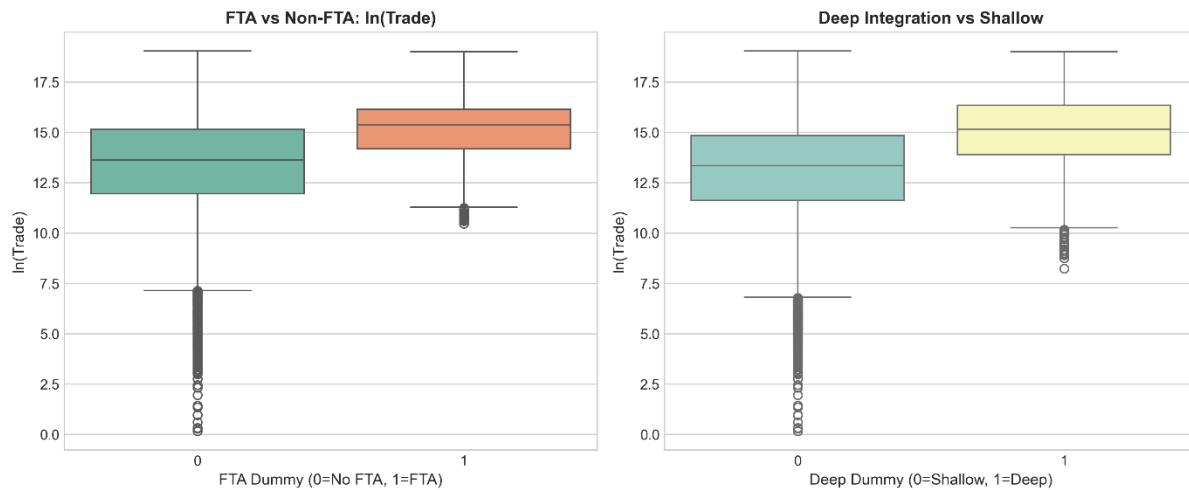


Graph 3.1 Predicted vs Actual (OLS & PPML)

Specification	fta_dummy	deep_dummy	Interaction	Key Findings
FTA × Asia	+1.00***	0.04	+	Modest but positive regional effect.
Deep × India	+1.05***	+0.13***	−1.02***	Deep FTAs with India underperform.
Lagged (t−1)	+0.31***	+1.05***	—	Lagged effects remain strong.

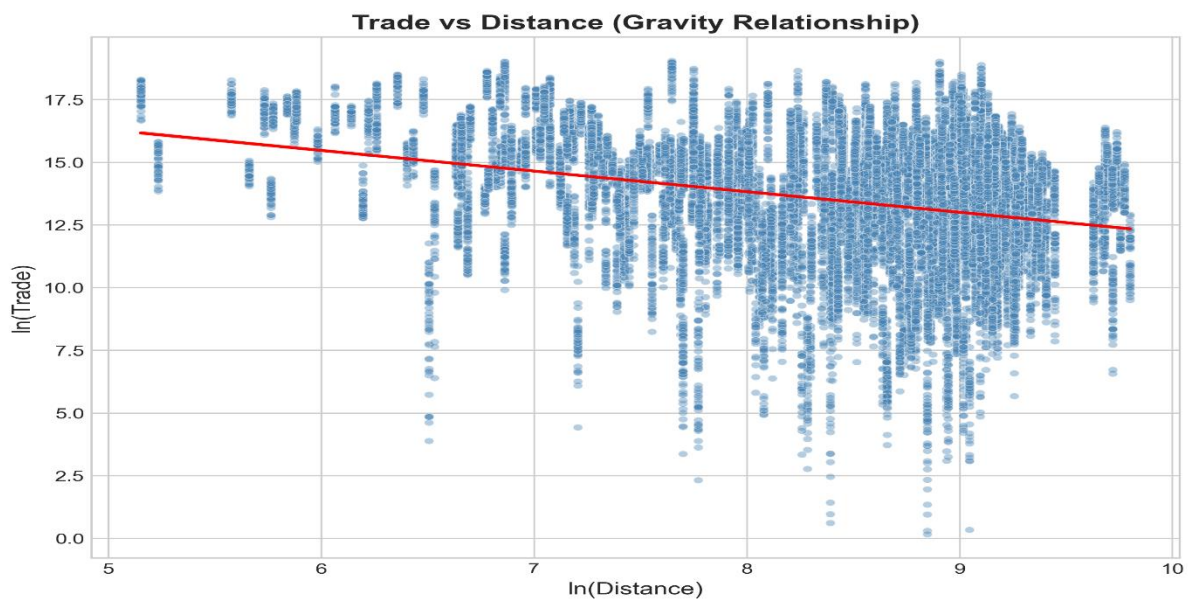
Excluding COVID (2020– 21)	+1.09***	0.02	—	Pandemic reduced trade volumes.
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Table 4.3 Interaction and Lagged Models

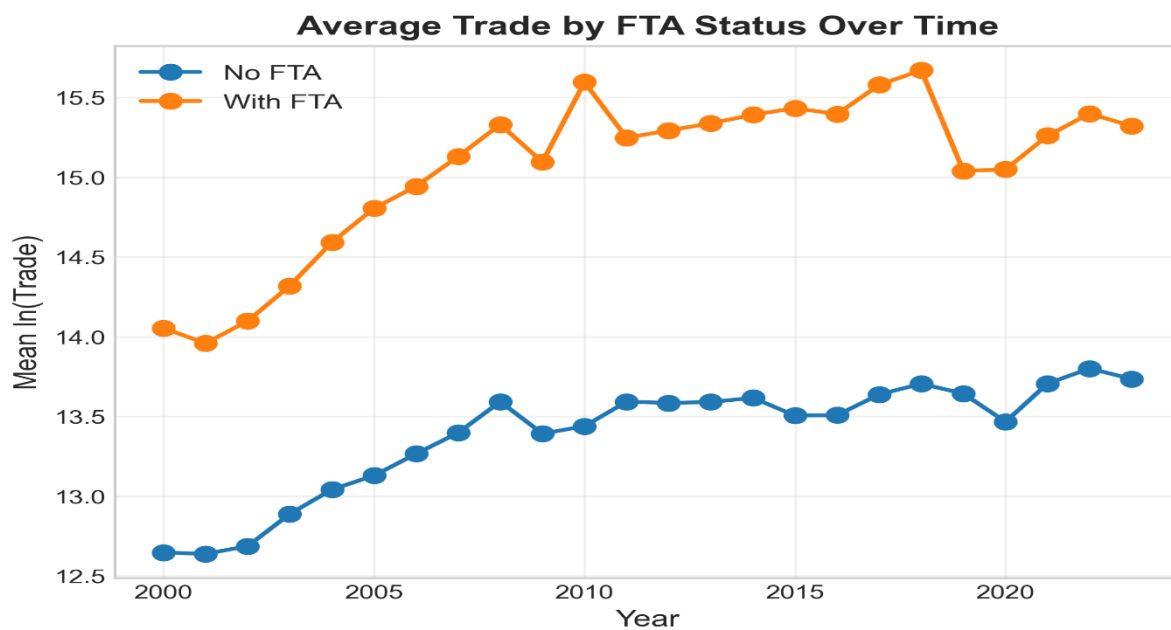


Graph 3.2 FTA and Deep dummy comparison

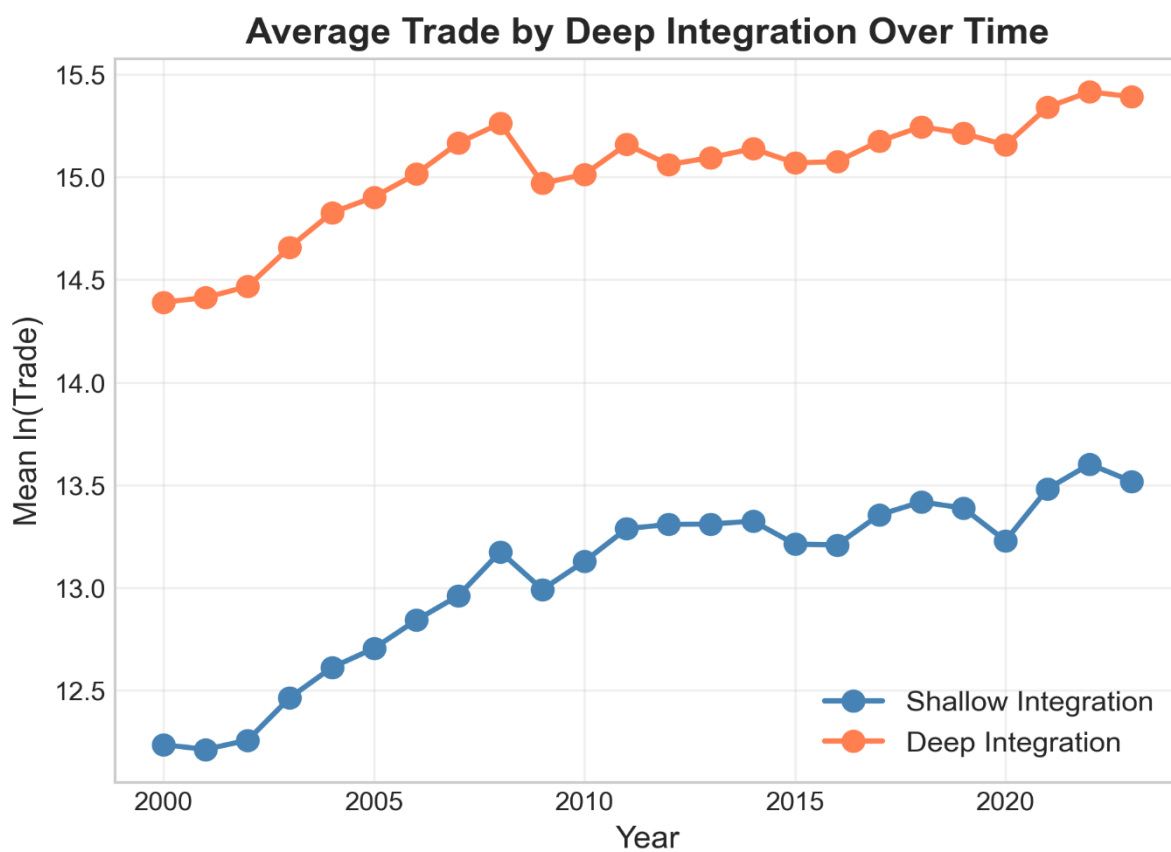
4.1 Interpretation:



Graph 4.1 FTA and Deep dummy comparison



Graph 4.2 FTA effect over time



Graph 4.3 Deep dummy effect over time

FTAs are found to be positively and significantly contributing to trade across models. The empirical results show that the presence of Free Trade Agreements increases the bilateral trade flows considerably. These findings corroborate the implications of both classical and new trade theories, which maintain that a reduction in tariffs and enhancement of market access provide a direct fillip to exchange between the partner economies. In both OLS and PPML specifications, the coefficient on the FTA dummy is positive and stands robust, thus proving that even without deep institutional commitments, Asia's regional FTAs have indeed successfully created more trade within the region.

In contrast, deep trade agreements yield weaker or even negative short-term effects, suggesting the presence of adjustment frictions and institutional inefficiencies. The negative sign on the deep integration variable in the PPML model indicates that while deeper commitments promise long-term benefits, they can initially impose compliance costs or reveal asymmetries in administrative capacity among partners. For developing economies with diverse regulatory systems, the deeper provisions will therefore result in uneven implementation and short-term trade distortions.

The negative interaction effect is highly significant at -1.02^{***} and shows that deep FTAs in India perform below expectations. This would point to structural and institutional constraints in the translation of deeper commitments into tangible gains in trade. When excluding pandemic years (2020–2021), the positive and significant effect of FTA is stronger, underlining the fact that external shocks obscured integration benefits. Compared with the European Union, the regional architecture in Asia is wanting in institutional depth: the harmonization of standards, dispute resolution mechanisms, and the enforcement credibility that underpins the EU common market success.

5. Conclusion and Policy Recommendations:

5.1 Summary:

This study assesses the prospect for a common market in Asia in the light of the role of Free Trade Agreements and depth of integration, using both OLS and PPML gravity models. However, the preliminary estimations done through OLS violated the two key assumptions of homoscedasticity and normality. Such violations were quite evident from the diagnostic tests undertaken using the Breusch-Pagan, White, and Jarque-Bera tests. Results from these diagnostics gave enough reason to reject the estimates through OLS in favour of the use of the Poisson Pseudo Maximum Likelihood estimator, which takes care of heteroscedasticity and nonlinearities present in trade data.

The positive and statistically significant estimates of the impact of FTAs on bilateral trade flows throughout Asia obtained using PPML, therefore, confirm that the FTAs have indeed been able to create regional trade. However, deeper trade agreements extending beyond tariff reduction to include legal, regulatory, and institutional harmonization are associated with mixed or even negative short-run effects, including those especially for the case of India. This then suggests that shallow liberalization has indeed paid dividends to Asia, but full benefits from deep integration have remained elusive due to structural, institutional, and implementation frictions.

5.2 Conclusion:

Asia's path towards a common market remains aspirational, rather than imminent. The empirical evidence from this study, based on a PPML gravity model with fixed effects, gives proof that while FTAs significantly boost bilateral flows of trade, the legal, regulatory, and institutional dimensions of deep integration have yielded either limited or even negative outcomes in the short run. OLS results were statistically invalid due to the presence of heteroscedasticity, reinforcing PPML's robustness and the credibility of these findings. Underlying India's underperformance in deep trade agreements is an implication towards the broader governance and structural gaps that lags the region's prospects for institutional convergence. Unlike the European Union, whose success stands on harmonized rules and enforceable commitments, integration in Asia has remained fragmentary, led more by economic pragmatism than by shared institutions. That would require solidification of the institutional architecture, policy coherence, and alignment of national reforms with regional ambitions if Asia is to go beyond an FTA network and take the leap toward a genuine common market-economic unity.

5.3 Policy Recommendations:

A common market for Asia is still an aspiration to be turned to reality. This study employs a PPML gravity model with fixed effects and finds that Free Trade Agreements have facilitated trade expansion among Asian economies, whereas more profound forms of integration-characterized by regulatory and institutional cooperation-have not produced equivalent benefits. The findings from OLS were deemed unreliable due to the presence of heteroskedasticity, thereby validating the use of PPML as a more accurate representation of trade relationships. India's subpar performance under deep trade agreements highlights the structural and policy challenges that numerous Asian economies encounter when attempting to implement complex trade agreements.

In contrast to the European Union, which benefits from shared institutions and consistent regulations, Asia's integration continues to function through a fragmented array of bilateral agreements with limited coordination. For the region to progress towards a true common market, it requires stronger institutions, clearer policy alignment, and enhanced trust among member nations. Only through such a coordinated initiative can Asia transform its expanding trade connections into a genuinely unified economic space.

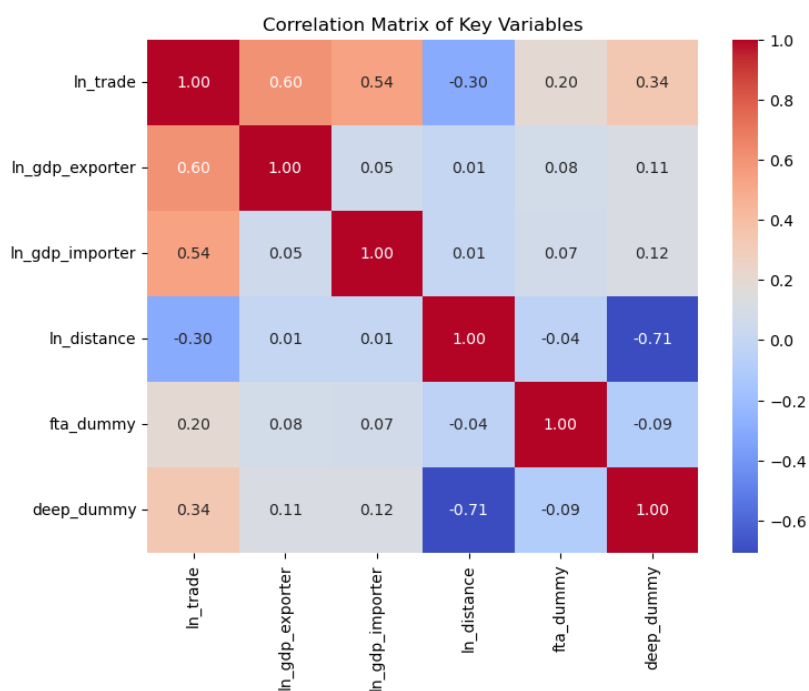
6. References:

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Appendix: (python outputs)



OLS Regression Results

```

=====
Dep. Variable:          ln_trade  R-squared:          0.726
Model:                  OLS  Adj. R-squared:         0.726
Method:                 Least Squares  F-statistic:      7152.
Date:                   Fri, 31 Oct 2025  Prob (F-statistic):    0.00
Time:                   01:34:02  Log-Likelihood:      -40277.
No. Observations:       23770  AIC:                   8.057e+04
Df Residuals:           23762  BIC:                   8.064e+04
Df Model:                7
Covariance Type:        HC1
=====

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=====
              coef    std err          z      P>|z|    [0.025    0.975]
-----
Intercept    -31.2777    0.295   -106.110    0.000   -31.855   -30.700
ln_gdp_exporter  1.0192    0.007   143.108    0.000    1.005    1.033
ln_gdp_importer  0.8961    0.007   127.279    0.000    0.882    0.910
ln_distance    -0.7980    0.015   -54.741    0.000   -0.827   -0.769
fta_dummy      1.0044    0.032    31.415    0.000    0.942    1.067
deep_dummy     0.0437    0.041     1.065    0.287   -0.037    0.124
eu_dummy       0.0533    0.036     1.467    0.142   -0.018    0.125
india_dummy    -0.2740    0.033   -8.375    0.000   -0.338   -0.210
=====

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=====
Omnibus:          1565.069  Durbin-Watson:          1.241
Prob(Omnibus):    0.000  Jarque-Bera (JB):        3498.881
Skew:             -0.428  Prob(JB):                 0.00
Kurtosis:         4.673  Cond. No.                 1.12e+03
=====

```

Notes:

[1] Standard Errors are heteroscedasticity robust (HC1)

[2] The condition number is large, 1.12e+03. This might indicate that there are strong multicollinearity or other numerical problems.

Generalized Linear Model Regression Results

```
=====
Dep. Variable:    trade_value_usd  No. Observations:    23770
Model:           GLM  Df Residuals:    23762
Model Family:    Poisson  Df Model:    7
Link Function:    log  Scale:    1.0000
Method:          IRLS  Log-Likelihood:    -3.7757e+10
Date:            Fri, 31 Oct 2025  Deviance:    7.5513e+10
Time:            01:37:23  Pearson chi2:    1.04e+11
No. Iterations:    8  Pseudo R-squ. (CS):    1.000
Covariance Type:    HC0
=====
```

```
=====
              coef  std err          z      P>|z|    [0.025    0.975]
-----+-----
const        -17.4282    0.310   -56.136    0.000   -18.037   -16.820
fta_dummy      0.5299    0.027   19.323    0.000    0.476    0.584
deep_dummy    -0.3579    0.042   -8.490    0.000   -0.441   -0.275
eu_dummy       0.2796    0.031    8.955    0.000    0.218    0.341
india_dummy   -0.4017    0.035  -11.474    0.000   -0.470   -0.333
ln_gdp_exporter  0.7255    0.007  109.217    0.000    0.713    0.739
ln_gdp_importer  0.6790    0.007   99.148    0.000    0.666    0.692
ln_distance   -0.6997    0.015  -46.234    0.000   -0.729   -0.670
=====
```

PPML WITH YEAR FIXED EFFECTS

Generalized Linear Model Regression Results

```
=====
Dep. Variable:    trade_value_usd  No. Observations:    23770
Model:           GLM  Df Residuals:    23739
Model Family:    Poisson  Df Model:    30
Link Function:    Log  Scale:    1.0000
Method:          IRLS  Log-Likelihood:    -3.6518e+10
Date:            Fri, 31 Oct 2025  Deviance:    7.3035e+10
Time:            01:40:53  Pearson chi2:    1.02e+11
No. Iterations:    11  Pseudo R-squ. (CS):    1.000
Covariance Type:    HC0
=====
```

```
=====
              coef  std err          z      P>|z|    [0.025    0.975]
-----+-----
const        -18.3949    0.311  -59.219    0.000   -19.004   -17.786
fta_dummy      0.6331    0.028   22.658    0.000    0.578    0.688
deep_dummy    -0.3619    0.042   -8.627    0.000   -0.444   -0.280
eu_dummy       0.2972    0.031    9.470    0.000    0.236    0.359
india_dummy   -0.3796    0.035  -10.933    0.000   -0.448   -0.312
ln_gdp_exporter  0.7476    0.007  111.769    0.000    0.734    0.761
ln_gdp_importer  0.7006    0.007  103.277    0.000    0.687    0.714
ln_distance   -0.6944    0.015  -46.047    0.000   -0.724   -0.665
year_2001     -0.0122    0.052   -0.234    0.815   -0.114    0.090
=====
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year_2002	-0.0559	0.053	-1.064	0.287	-0.159	0.047
year_2003	-0.1259	0.054	-2.345	0.019	-0.231	-0.021
year_2004	-0.1352	0.055	-2.443	0.015	-0.244	-0.027
year_2005	-0.1111	0.056	-1.977	0.048	-0.221	-0.001
year_2006	-0.0890	0.056	-1.586	0.113	-0.199	0.021
year_2007	-0.1447	0.056	-2.561	0.010	-0.255	-0.034
year_2008	-0.1499	0.056	-2.678	0.007	-0.260	-0.040
year_2009	-0.2880	0.055	-5.245	0.000	-0.396	-0.180
year_2010	-0.2362	0.055	-4.325	0.000	-0.343	-0.129
year_2011	-0.2358	0.055	-4.326	0.000	-0.343	-0.129
year_2012	-0.2557	0.054	-4.715	0.000	-0.362	-0.149
year_2013	-0.3020	0.054	-5.586	0.000	-0.408	-0.196
year_2014	-0.3376	0.054	-6.282	0.000	-0.443	-0.232
year_2015	-0.3776	0.053	-7.096	0.000	-0.482	-0.273
year_2016	-0.4233	0.053	-8.010	0.000	-0.527	-0.320
year_2017	-0.3804	0.053	-7.131	0.000	-0.485	-0.276
year_2018	-0.4112	0.053	-7.766	0.000	-0.515	-0.307
year_2019	-0.4797	0.054	-8.825	0.000	-0.586	-0.373
year_2020	-0.5261	0.055	-9.556	0.000	-0.634	-0.418
year_2021	-0.4942	0.056	-8.828	0.000	-0.604	-0.384
year_2022	-0.3819	0.056	-6.867	0.000	-0.491	-0.273
year_2023	-0.5071	0.055	-9.244	0.000	-0.615	-0.400

PPML WITH DYADIC AND YEAR FIXED EFFECTS

Generalized Linear Model Regression Results

Dep. Variable:	trade_value_usd	No. Observations:	23770
Model:	GLM	Df Residuals:	22751
Model Family:	Poisson	Df Model:	1018
Link Function:	Log	Scale:	1.0000
Method:	IRLS	Log-Likelihood:	-2.9473e+09
Date:	Fri, 31 Oct 2025	Deviance:	5.8942e+09
Time:	01:44:07	Pearson chi2:	5.97e+09
No. Iterations:	20	Pseudo R-squ. (CS):	1.000
Covariance Type:	HC0		

KEY COEFFICIENTS OF INTEREST

fta_dummy	: -0.0006 (SE: 0.0168, p-value: 0.9714)
deep_dummy	: 0.0666 (SE: 0.0242, p-value: 0.0059)
ln_gdp_exporter	: 0.5224 (SE: 0.0128, p-value: 0.0000)
ln_gdp_importer	: 0.5419 (SE: 0.0157, p-value: 0.0000)

NORMALITY TESTS FOR OLS RESIDUALS

Jarque-Bera Test:

Statistic:	3498.8809
p-value:	0.0000
Skewness:	-0.4284
Kurtosis:	4.6729

Result: Reject normality ($\alpha=0.05$)

Shapiro-Wilk Test:

Statistic: 0.9876

p-value: 0.0000

Result: Reject normality ($\alpha=0.05$)

Residual Statistics:

Mean: -0.000000

Std Dev: 1.3172

Min: -9.8191

Max: 4.6729

Interaction: FTA $\times$ Asia (OLS)

OLS Regression Results

```
=====
Dep. Variable:      ln_trade  R-squared:      0.726
Model:              OLS      Adj. R-squared:    0.726
Method:              Least Squares  F-statistic:  7152.
Date:                Fri, 31 Oct 2025  Prob (F-statistic): 0.00
Time:                02:04:33  Log-Likelihood:  -40277.
No. Observations:    23770  AIC:                8.057e+04
Df Residuals:        23762  BIC:                8.064e+04
Df Model:             7
Covariance Type:     HC1
=====
```

```
=====
              coef  std err      z  P>|z|  [0.025  0.975]
-----
Intercept    -31.2777   0.295 -106.110  0.000  -31.855  -30.700
ln_gdp_exporter  1.0192   0.007  143.108  0.000   1.005   1.033
ln_gdp_importer  0.8961   0.007  127.279  0.000   0.882   0.910
ln_distance     -0.7980   0.015  -54.741  0.000  -0.827  -0.769
fta_dummy       1.0044   0.032   31.415  0.000   0.942   1.067
fta_x_asia     7.099e-16  7.28e-17  9.749   0.000  5.67e-16  8.53e-16
deep_dummy      0.0437   0.041   1.065   0.287  -0.037   0.124
eu_dummy        0.0533   0.036   1.467   0.142  -0.018   0.125
india_dummy     -0.2740   0.033  -8.375   0.000  -0.338  -0.210
=====
```

```
=====
Omnibus:          1565.069  Durbin-Watson:      1.241
Prob(Omnibus):    0.000  Jarque-Bera (JB):    3498.881
Skew:             -0.428  Prob(JB):            0.00
Kurtosis:         4.673  Cond. No.            2.65e+18
=====
```

Notes:

[1] Standard Errors are heteroscedasticity robust (HC1)

[2] The smallest eigenvalue is 5.11e-30. This might indicate that there are strong multicollinearity problems or that the design matrix is singular.

Interaction: Deep $\times$ India (OLS)

OLS Regression Results

Dep. Variable: ln_trade R-squared: 0.727
 Model: OLS Adj. R-squared: 0.727
 Method: Least Squares F-statistic: 6255.
 Date: Fri, 31 Oct 2025 Prob (F-statistic): 0.00
 Time: 02:04:33 Log-Likelihood: -40257.
 No. Observations: 23770 AIC: 8.053e+04
 Df Residuals: 23761 BIC: 8.060e+04
 Df Model: 8
 Covariance Type: HC1

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-31.3180	0.294	-106.354	0.000	-31.895	-30.741
ln_gdp_exporter	1.0175	0.007	143.091	0.000	1.004	1.031
ln_gdp_importer	0.8944	0.007	127.002	0.000	0.881	0.908
ln_distance	-0.7837	0.015	-53.260	0.000	-0.813	-0.755
fta_dummy	1.0501	0.033	32.023	0.000	0.986	1.114
deep_dummy	0.1268	0.042	3.013	0.003	0.044	0.209
deep_x_india	-1.0193	0.128	-7.962	0.000	-1.270	-0.768
eu_dummy	0.0036	0.037	0.097	0.923	-0.068	0.076
india_dummy	-0.2110	0.033	-6.303	0.000	-0.277	-0.145
Omnibus:	1553.772	Durbin-Watson:	1.241			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	3471.773			
Skew:	-0.426	Prob(JB):	0.00			
Kurtosis:	4.668	Cond. No.	1.12e+03			

Notes:

[1] Standard Errors are heteroscedasticity robust (HC1)

[2] The condition number is large, 1.12e+03. This might indicate that there are strong multicollinearity or other numerical problems.

Lagged FTA / Deep (OLS)

OLS Regression Results

=====

Dep. Variable:	ln_trade	R-squared:	0.730
Model:	OLS	Adj. R-squared:	0.730
Method:	Least Squares	F-statistic:	5386.
Date:	Fri, 31 Oct 2025	Prob (F-statistic):	0.00
Time:	02:04:33	Log-Likelihood:	-38379.
No. Observations:	22778	AIC:	7.678e+04
Df Residuals:	22768	BIC:	7.686e+04
Df Model:	9		
Covariance Type:	HC1		

=====

	coef	std err	z	P> z	[0.025	0.975]

Intercept	-32.0371	0.304	-105.394	0.000	-32.633	-31.441
ln_gdp_exporter	1.0341	0.007	141.496	0.000	1.020	1.048
ln_gdp_importer	0.9094	0.007	126.272	0.000	0.895	0.924
ln_distance	-0.8015	0.015	-53.897	0.000	-0.831	-0.772

fta_dummy	0.3095	0.088	3.526	0.000	0.137	0.482
fta_lag1	0.7373	0.092	7.992	0.000	0.556	0.918
deep_dummy	1.0508	0.191	5.504	0.000	0.677	1.425
deep_lag1	-1.0414	0.194	-5.356	0.000	-1.422	-0.660
eu_dummy	0.0828	0.037	2.222	0.026	0.010	0.156
india_dummy	-0.2831	0.034	-8.427	0.000	-0.349	-0.217
=====						
Omnibus:	1475.778	Durbin-Watson:	0.206			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	3373.425			
Skew:	-0.415	Prob(JB):	0.00			
Kurtosis:	4.693	Cond. No.	1.24e+03			
=====						

Notes:

[1] Standard Errors are heteroscedasticity robust (HC1)

[2] The condition number is large, 1.24e+03. This might indicate that there are strong multicollinearity or other numerical problems.

Trade growth regression (OLS)

OLS Regression Results

Dep. Variable:	trade_growth	R-squared:	0.002
Model:	OLS	Adj. R-squared:	0.002
Method:	Least Squares	F-statistic:	10.37
Date:	Fri, 31 Oct 2025	Prob (F-statistic):	4.71e-13
Time:	02:04:34	Log-Likelihood:	-10106.
No. Observations:	22778	AIC:	2.023e+04
Df Residuals:	22770	BIC:	2.029e+04
Df Model:	7		
Covariance Type:	HC1		
	coef	std err	z P> z [0.025 0.975]
Intercept	0.3742	0.116	3.214 0.001 0.146 0.602
fta_lag1	-0.0006	0.007	-0.089 0.929 -0.015 0.013
deep_lag1	-0.0369	0.008	-4.378 0.000 -0.053 -0.020
ln_gdp_exporter	-0.0051	0.002	-2.227 0.026 -0.010 -0.001
ln_gdp_importer	-0.0040	0.003	-1.548 0.122 -0.009 0.001
ln_distance	-0.0080	0.004	-1.868 0.062 -0.016 0.000
eu_dummy	0.0044	0.008	0.579 0.562 -0.011 0.019
india_dummy	0.0373	0.008	4.429 0.000 0.021 0.054
Omnibus:	5962.910	Durbin-Watson:	2.437
Prob(Omnibus):	0.000	Jarque-Bera (JB):	549254.416
Skew:	-0.019	Prob(JB):	0.00
Kurtosis:	27.057	Cond. No.	1.13e+03

Notes:

[1] Standard Errors are heteroscedasticity robust (HC1)

[2] The condition number is large, 1.13e+03. This might indicate that there are

strong multicollinearity or other numerical problems.

OLS base with dyad-clustered SEs

OLS Regression Results

Dep. Variable:	ln_trade	R-squared:	0.726
Model:	OLS	Adj. R-squared:	0.726
Method:	Least Squares	F-statistic:	449.5
Date:	Fri, 31 Oct 2025	Prob (F-statistic):	2.39e-302
Time:	02:04:34	Log-Likelihood:	-40277.
No. Observations:	23770	AIC:	8.057e+04
Df Residuals:	23762	BIC:	8.064e+04
Df Model:	7		
Covariance Type:	cluster		

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-31.2777	1.190	-26.283	0.000	-33.610	-28.945
ln_gdp_exporter	1.0192	0.030	34.405	0.000	0.961	1.077
ln_gdp_importer	0.8961	0.029	31.404	0.000	0.840	0.952
ln_distance	-0.7980	0.060	-13.237	0.000	-0.916	-0.680
fta_dummy	1.0044	0.118	8.478	0.000	0.772	1.237
deep_dummy	0.0437	0.158	0.277	0.782	-0.265	0.353
eu_dummy	0.0533	0.148	0.359	0.720	-0.238	0.344
india_dummy	-0.2740	0.138	-1.991	0.046	-0.544	-0.004

Omnibus:	1565.069	Durbin-Watson:	0.195
Prob(Omnibus):	0.000	Jarque-Bera (JB):	3498.881
Skew:	-0.428	Prob(JB):	0.00
Kurtosis:	4.673	Cond. No.	1.12e+03

Notes:

[1] Standard Errors are robust to cluster correlation (cluster)

[2] The condition number is large, 1.12e+03. This might indicate that there are strong multicollinearity or other numerical problems.

PPML baseline (clustered by dyad)

Generalized Linear Model Regression Results

Dep. Variable:	trade_value_usd	No. Observations:	23770
Model:	GLM	Df Residuals:	23762
Model Family:	Poisson	Df Model:	7
Link Function:	Log	Scale:	1.0000
Method:	IRLS	Log-Likelihood:	-3.7757e+10
Date:	Fri, 31 Oct 2025	Deviance:	7.5513e+10
Time:	02:04:34	Pearson chi2:	1.04e+11
No. Iterations:	8	Pseudo R-squ. (CS):	1.000
Covariance Type:	cluster		

	coef	std err	z	P> z	[0.025	0.975]
const	-17.4282	1.219	-14.297	0.000	-19.817	-15.039
fta_dummy	0.5299	0.088	6.016	0.000	0.357	0.702
deep_dummy	-0.3579	0.148	-2.412	0.016	-0.649	-0.067
eu_dummy	0.2796	0.131	2.134	0.033	0.023	0.537
india_dummy	-0.4017	0.137	-2.935	0.003	-0.670	-0.133
ln_gdp_exporter	0.7255	0.026	28.439	0.000	0.676	0.776
ln_gdp_importer	0.6790	0.026	25.881	0.000	0.628	0.730
ln_distance	-0.6997	0.060	-11.566	0.000	-0.818	-0.581

PPML with year fixed effects (clustered by dyad)

Generalized Linear Model Regression Results

```

=====
Dep. Variable:      trade_value_usd  No. Observations:      23770
Model:              GLM  Df Residuals:      23739
Model Family:       Poisson  Df Model:      30
Link Function:      Log  Scale:      1.0000
Method:             IRLS  Log-Likelihood:  -3.6518e+10
Date:              Fri, 31 Oct 2025  Deviance:      7.3035e+10
Time:              02:04:34  Pearson chi2:      1.02e+11
No. Iterations:     7  Pseudo R-squ. (CS):      1.000
Covariance Type:    cluster
=====

```

	coef	std err	z	P> z	[0.025	0.975]
const	-18.3949	1.269	-14.490	0.000	-20.883	-15.907
fta_dummy	0.6331	0.090	7.036	0.000	0.457	0.809
deep_dummy	-0.3619	0.150	-2.408	0.016	-0.656	-0.067
eu_dummy	0.2972	0.133	2.229	0.026	0.036	0.559
india_dummy	-0.3796	0.139	-2.728	0.006	-0.652	-0.107
ln_gdp_exporter	0.7476	0.027	28.041	0.000	0.695	0.800
ln_gdp_importer	0.7006	0.027	25.876	0.000	0.648	0.754
ln_distance	-0.6944	0.061	-11.299	0.000	-0.815	-0.574
yr_2001	-0.0122	0.007	-1.867	0.062	-0.025	0.001
yr_2002	-0.0559	0.011	-5.033	0.000	-0.078	-0.034
yr_2003	-0.1259	0.018	-6.879	0.000	-0.162	-0.090
yr_2004	-0.1352	0.023	-5.793	0.000	-0.181	-0.089
yr_2005	-0.1111	0.024	-4.625	0.000	-0.158	-0.064
yr_2006	-0.0890	0.026	-3.419	0.001	-0.140	-0.038
yr_2007	-0.1447	0.030	-4.791	0.000	-0.204	-0.085
yr_2008	-0.1499	0.033	-4.481	0.000	-0.216	-0.084
yr_2009	-0.2880	0.030	-9.518	0.000	-0.347	-0.229
yr_2010	-0.2362	0.030	-7.768	0.000	-0.296	-0.177
yr_2011	-0.2358	0.034	-7.012	0.000	-0.302	-0.170
yr_2012	-0.2557	0.034	-7.480	0.000	-0.323	-0.189
yr_2013	-0.3020	0.035	-8.594	0.000	-0.371	-0.233
yr_2014	-0.3376	0.035	-9.661	0.000	-0.406	-0.269
yr_2015	-0.3776	0.034	-11.218	0.000	-0.444	-0.312
yr_2016	-0.4233	0.034	-12.394	0.000	-0.490	-0.356

yr_2017	-0.3804	0.037	-10.244	0.000	-0.453	-0.308
yr_2018	-0.4112	0.039	-10.475	0.000	-0.488	-0.334
yr_2019	-0.4797	0.039	-12.305	0.000	-0.556	-0.403
yr_2020	-0.5261	0.038	-13.854	0.000	-0.601	-0.452
yr_2021	-0.4942	0.043	-11.386	0.000	-0.579	-0.409
yr_2022	-0.3819	0.045	-8.522	0.000	-0.470	-0.294
yr_2023	-0.5071	0.044	-11.559	0.000	-0.593	-0.421

OLS excluding pandemic years (2020–2021)

OLS Regression Results

```

Dep. Variable:      ln_trade  R-squared:      0.728
Model:              OLS      Adj. R-squared:    0.728
Method:             Least Squares  F-statistic:  6635.
Date:               Fri, 31 Oct 2025  Prob (F-statistic): 0.00
Time:               02:04:34  Log-Likelihood:  -36883.
No. Observations:   21786  AIC:               7.378e+04
Df Residuals:       21778  BIC:               7.385e+04
Df Model:           7
Covariance Type:    HC1

```

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-31.3741	0.305	-102.815	0.000	-31.972	-30.776
ln_gdp_exporter	1.0201	0.007	138.542	0.000	1.006	1.035
ln_gdp_importer	0.9009	0.007	123.064	0.000	0.887	0.915
ln_distance	-0.7991	0.015	-52.908	0.000	-0.829	-0.769
fta_dummy	1.0884	0.035	30.714	0.000	1.019	1.158
deep_dummy	0.0151	0.044	0.346	0.729	-0.071	0.101
eu_dummy	0.0519	0.038	1.353	0.176	-0.023	0.127
india_dummy	-0.2772	0.033	-8.287	0.000	-0.343	-0.212

```

Omnibus:           1354.142  Durbin-Watson:      0.212
Prob(Omnibus):     0.000  Jarque-Bera (JB):    2911.423
Skew:              -0.417  Prob(JB):            0.00
Kurtosis:          4.584  Cond. No.            1.11e+03

```

Notes:

[1] Standard Errors are heteroscedasticity robust (HC1)

[2] The condition number is large, 1.11e+03. This might indicate that there are strong multicollinearity or other numerical problems.

